

SAMUEL GRAYSON

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EDUCATION

NORTHWESTERN UNIVERSITY

Bachelor of Science: Computer Engineering

Dean's List Fall 2024, Winter 2025, Spring 2025, High Honors Fall 2024

Relevant Coursework: Data Structures and Algorithms (DSA), Computer Systems, Advanced Digital Design

Evanston, IL

Candidate, Jun 2027

GPA: 3.90/4.00

SKILLS

Programing Languages: C, C++, Java, JavaScript, Python, Scamper (Scheme), SQL, Racket, Verilog, SystemVerilog

Web Development: HTML, CSS, Bootstrap, NextJS, REST API Design and Development, WebSockets, Flask, gUnicorn, Nginx

Software: Autodesk Inventor, CAD, Altium, KiCad EDA, EasyEDA, Microsoft Office 365, Adobe Creative Suite, Waveforms

Operating Systems: IBM VSE/ESA, IBM Z/OS, Linux, Microsoft Windows, MacOS

Languages: English, Russian

Other: Arduino, ESXi, Git, PlatformIO, DBeaver, TIBCO Spotfire, PostgreSQL, SQLite, ETC Eos, QLab, MagicQ, Dante

EXPERIENCE

UNITED AIRLINES

Software Engineering Intern

Chicago, IL

Jun 2025 – Sep 2025

- Developed a full stack Flask application enabling seamless communication between flight crews and the airline's hotel operations team using Python, JavaScript, WebSockets, RESTful APIs, Twilio for SMS integration
- Implemented generative AI (LLMs) to automatically classify and route plain-text flight crew inquiries to the appropriate hotel scheduler, while performing initial data collection to streamline workflows and significantly reduce ticket resolution time
- Created a new dashboard in TIBCO Spotfire using SQL and Amazon Redshift to quickly identify which passengers on a given flight can be rerouted with minimal impact in the event of an oversale situation

VAK SUSTAINABLE COMPUTING LAB

Undergraduate Researcher

Evanston, IL

Jan 2025 - Current

- Built and programmed a multispectral imaging system using Raspberry Pis, ESP32s, and multiple cameras with spectral filters

THE ELECTRIC TUBE COMPANY

Founder

Bexley, OH

Oct 2020 - Current

- Imported vintage vacuum and nixie tube equipment from Europe and Central Asia for resale in America (eCommerce)
- Developed strategic business partnerships with 10+ Russian speaking suppliers overseas
- Designed custom PCBs for computer-controlled nixie tube displays

BEXLEY CITY SCHOOLS

Lighting and Sound Technician

Bexley, OH

Apr 2021 - Jun 2023

- Oversaw a 10-person team and managed all lighting, sound, and video systems in Bexley City School's 3 theatrical venues
- Collaborated with vendors and stakeholders to organize >\$100,000 in capital improvements in lighting and sound equipment

PROJECTS

LEAKLOGIK VALVES

Jan 2024 - Current

- Secured \$1500 in seed funding to develop a patent pending microprocessor-controlled leak-sensing toilet shutoff valve
- Conducted extensive product-market fit and patent research, informing design decisions and guiding strategy for IP development
- Established business relations directly with factories in China to source key components from

IEEE CHIP DESIGN TEAM (NU)

Jan 2025 - May 2025

- Designed a convolutional filter in RTL for use in the team's sign language recognition (computer vision) FPGA project

ADDITIONAL

Home Lab: Operate a variety of servers and networking equipment as well as an IBM MP3000 mainframe

Clubs: President of the Northwestern Russian Club, Resident at The Garage (Northwestern), NU Aviation Club Treasurer

Other: Received a fully funded U.S. Department of State ECA scholarship, CLS, to study Russian abroad in Kyrgyzstan (2024)